Oligopoly

Answer key with worked solutions

For P = A - bQ and identical MC = c, symmetric Cournot duopoly satisfies A - 3bq = c, so each firm produces q = (A - c)/(3b) and total Q = 2q. Standard identical-product Bertrand gives P = MC. With no fixed costs, profit per firm is π = (P - MC)q.

Source: Oligopoly (1).pdf

#	Answer	Work / explanation
1	**d - $5**	C(q) = 5q, so MC = 5. With identical products and costs, Bertrand price competition gives P = MC = $5.
2	**a**	Holding demand and costs fixed, monopoly restricts output most, Cournot duopoly supplies more, and perfect competition supplies the most. With downward-sloping demand: P(monopoly) > P(Cournot) > P(perfect competition).
3	**c - Monopolistic competition**	
4	**b - $40**	Firm A has MR_A = 100 - 2qA - qB. Set MR_A = MC = 10. By symmetry qA = qB = q, so 100 - 3q = 10 and q = 30. Total Q = 60, giving P = 100 - 60 = $40.
5	**c, d**	Oligopoly and collusion. A few large suppliers dominate the market, and the alleged agreement to restrict production and raise price is collusion. The scenario does not describe regulatory capture.
6	**b - 50**	MC = 10. Bertrand equilibrium gives P = MC = 10. Substitute into market demand: Q = 60 - 10 = 50.
7	**b - $1,600**	Cournot output per firm is q = (180 - 60)/3 = 40. Total Q = 80 and P = 180 - 80 = $100. With no fixed costs, each firm's profit is π = (P - MC)q = (100 - 60)(40) = $1,600.
8	**c**	A monopolistically competitive firm with few substitutes faces less elastic demand and can sustain a larger markup. At its optimum, L = (P - MC)/P = 1/|ε|; a smaller elasticity magnitude implies a higher Lerner index.
9	**e - None**	The correct order is Perfect Competition > Cournot > Monopoly. For P = A - bQ and common MC = c: Q(PC) = (A - c)/b, Q(Cournot) = 2(A - c)/(3b), and Q(monopoly) = (A - c)/(2b). None of a–d lists this order.
10	**a**	Intended textbook ordering: Monopoly > Monopolistic Competition > Perfect Competition. With comparable conditions, more substitutes and competition reduce the price-cost markup; perfect competition gives P = MC.
11	**b, c**	Bertrand: P = MC = 20, so Q = 60 - 20 = 40. Cournot total output is 2(60 - 20)/3 = 80/3 ≈ 26.67, which is lower. Perfect competition also gives Q = 40, so Bertrand output is not higher than competitive output.

#	Answer	Work / explanation
12	**E - None** **q = 100/3 ≈ 33.33**	$MR_1 = 120 - 2q_1 - q_2 = 20$. Symmetry gives $q_1 = q_2 = q$, so $120 - 3q = 20$. Therefore each firm produces $q = 100/3 \approx 33.33$, which is not listed.
13	**d - 60**	Bertrand gives $P = MC = 50$. Set $350 - 5Q = 50$, so $5Q = 300$ and total market quantity $Q = 60$.
14	**A - $20**	The Bertrand equilibrium is $P = MC = \$20$. At a common price above $20, either firm can profitably undercut the other. Pricing below $20 would cause losses on every unit sold.
15	**first a, b (1st and 3rd statements)**	Select “Equal to marginal cost” and “Equal to the price in perfect competition.” Standard Bertrand gives $P = MC = P_{PC}$. The source repeats the label a; the correct statements are the first and third.
16	**b - 30**	$MC = 70$ and inverse demand is $P = 160 - Q$. Cournot output per firm is $q = (160 - 70)/3 = 30$. Thus Firm 2 produces $q_2 = 30$. The question’s reference to Q1 for Firm 2 is a subscript typo.
17	**a - 40**	$MC = 20$. Bertrand gives $P = MC = 20$, so total market quantity is $Q = 60 - 20 = 40$.
18	**c - 26.67**	Cournot output per firm is $q = (60 - 20)/3 = 40/3$. Total market quantity is $Q = 2q = 80/3 \approx 26.67$. Cournot is quantity competition; the question’s parenthetical “price” is a typo.
19	**E - None** **q = 80/3 ≈ 26.67**	MR_A = 100 - 2qA - qB = 20. Symmetry gives qA = qB = q, so 100 - 3q = 20. Each firm produces q = 80/3 ≈ 26.67, which is not listed.
20	**b - 55**	$C(q) = 5q$, so $MC = 5$. Bertrand price is $P = MC = 5$. Total market quantity is $Q = 60 - 5 = 55$.
21	**c, d**	Oligopoly and collusion, as in Question 5. The few major suppliers form an oligopoly; if the alleged agreement to limit output and increase price is true, their conduct is collusion.